Phase 5: Project Development

1. Overview

The Project Development phase involved the actual coding and implementation of the plans defined in the previous phases.

Note on Repository Structure: To maintain compatibility with cloud deployment platforms (like Heroku or Render), the source code is kept in the root directory rather than being moved into this folder.

2. Key Components Developed

Data Engineering (`notebooks/`)

Jupyter notebooks were used extensively to clean the Kaggle dataset. Code was written to map categorical variables (like "Yes"/"No") into numerical binaries (1/0) and to split the data into training and testing sets.

Model Training (`train_model.py`)

This script automates the training pipeline. It reads the preprocessed data, fits the XGBoost classifier, and serializes (pickles) the trained model into the `app/models/` directory for production use.

The Application Core (`app/app.py` & `run.py`)

The Flask framework was implemented to serve the application. * `run.py` acts as the entry point. * `app.py` handles the logic: receiving form data via POST requests, formatting the data into a NumPy array, passing it to the loaded model, and rendering the result template.

Frontend (`app/templates/` & `app/static/`)

HTML templates were built using Jinja2 syntax to dynamically display the prediction results and probabilities returned by the Flask backend.